B.Sc. Engg. SWE 1st Semester

31 october 2024 (Morning)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 2 HOURS

WINTER SEMESTER, 2023-2024
FULL MARKS: 120

SWE 4101: Introduction to Software Engineering

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) A computer system has a memory hierarchy from registers to main memory and secondary storage. Draw and explain the memory hierarchy according to access speed, cost and storage capacity. — 6 (CO1) (PO1)

   b) A number of factors related to the CPU have effect on the overall speed and performance of a computer. Briefly describe these factors. — 7 (CO1) (PO1)

   c) Compare the five generations of computers in details based on their hardware, software, computing characteristics, physical appearance, and applications. — 10 (CO1) (PO1)

   d) Briefly explain the developments in computer technology starting from a simple calculating machine till the first computer was developed. — 7 (CO1) (PO1)

2. a) Explain the steps of the scenario from typing an URL on the browser to receiving a web page in response? DNS — 8 (CO1) (PO1)

   b) Define the term protocol. Briefly explain the functionalities of different protocols used in computer networking. — 7 (CO1) (PO1)

   c) Differentiate between Client-Server and Peer-to-Peer Network. When should you choose a Peer-to-Peer Network over a Client-Server Network? — 4 + 3 (CO1) (PO1)

   d) Program Development Life Cycle (PDLC) is a process that software engineers and product teams use to develop quality software. Explain the phases of PDLC. — 8 (CO1) (PO1)

3. a) Differentiate between compiler, interpreter, and assembler. — 5 (CO1) (PO1)

   b) There are four phases to convert a C program source code to an executable. Explain these phases. PP → CO → Linking — 10 (CO1) (PO1)

   c) Fibonacci Series is a series of numbers in which each number is the sum of the two preceding numbers. For example, Fibonacci series upto 7 terms is — 1, 1, 2, 3, 5, 8, 13. Write an algorithm and flowchart to print the fibonacci series till the last term is less than 1000. — 10 (CO2) (PO2)

   d) Differentiate between syntax error and runtime error with examples. — 5 (CO1) (PO1)

4. a) Explain the significance of face value and position value of a number with an example. Perform the following number conversion: 3 + 6 (CO1) (PO1)

   i $(1011.1001)_2 = (?)_{10}$

   ii $(0.865)_{10} = (?)_2$

   iii $(yourstudentID)_{10} = (?)_{16}$

   b) Why 2's complement is advantageous over 1's complement for representing signed binary numbers in a computer system? Represent the following as 8-bit numbers in — Signed Magnitude representation, Signed 1's complement representation, and Signed 2's complement representation: 2 + 6 (CO1) (PO1)

   i 55

   ii 34

   iii 67

   c) Let, $m = (313)_4$ and $n = (322)_4$. Find $(m + n)_4$. 7 (CO1) (PO1)

   d) List and describe the three commonly used binary coding schemes. 6 (CO1) (PO1)